

ITSOLERA Pvt. Ltd.

Machine Learning Internship Project

Project Title: Adaptive Synthetic Data Augmentation Toolkit

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1. Introduction:

In modern machine learning applications, access to balanced and high-quality data is often a challenge. Real-world datasets, particularly in domains like finance and healthcare, frequently suffer from class imbalance, resulting in poor model performance on critical minority classes, such as fraud detection.

This project presents an **Adaptive Synthetic Data Augmentation Toolkit**, designed to dynamically generate synthetic data based on model weaknesses. Unlike static augmentation techniques, this approach employs a **closed-loop feedback mechanism** to iteratively improve model performance.

2. Problem Statement:

Traditional data augmentation methods are static and do not adapt to the model's weaknesses. Highly imbalanced datasets cause models to perform well on majority classes while failing on minority classes.

The goal of this project is to develop a system that: - Identifies model errors, especially false negatives - Generates targeted synthetic data - Retrains the model iteratively - Enhances performance on minority classes

3. Dataset Description:

project uses a real-world credit card fraud detection dataset: - Training samples: 227,845 - Features: 30 - Class distribution: - Class 0 (Normal): 227,451 - Class 1 (Fraud): 394

The extreme imbalance requires an intelligent augmentation approach to improve minority class detection.

4. Methodology:

4.1 Baseline Model

A Random Forest Classifier was trained on the dataset using stratified splits: - Recall (Class 1): 0.7449 - F1-score (Class 1): 0.84

This baseline highlighted the challenge in detecting fraud cases.

4.2 Error Analysis

The system identifies **false negatives**, cases where fraudulent transactions are classified as normal, which are critical for model improvement.

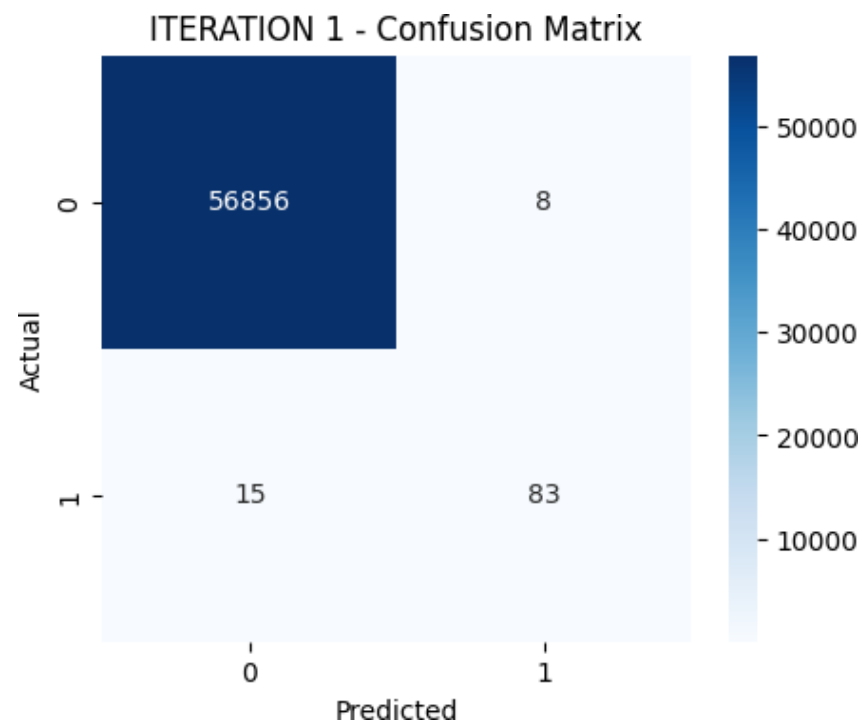
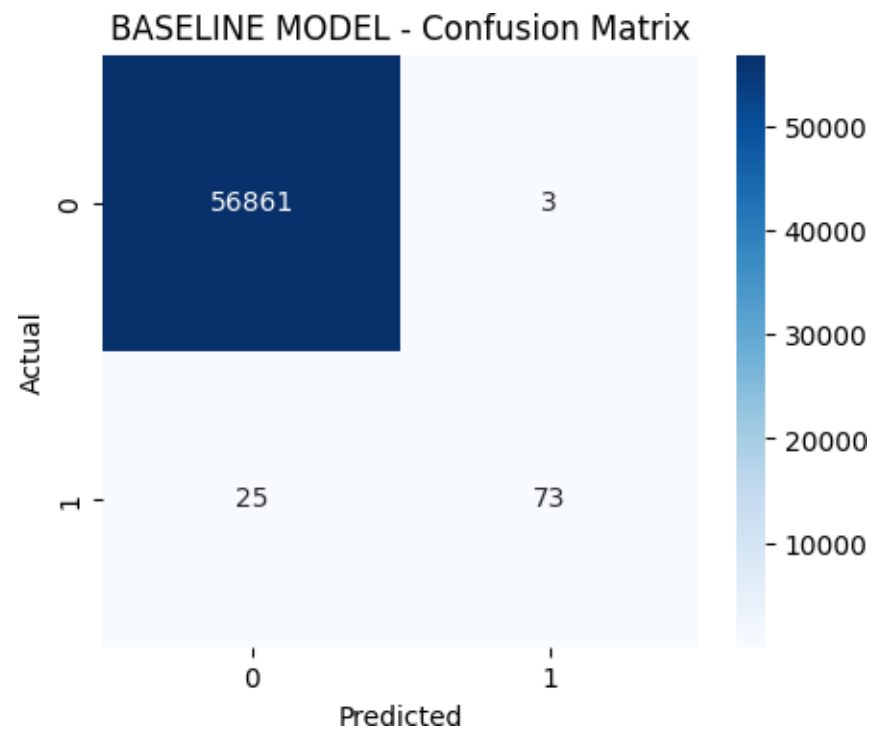
4.3 Synthetic Data Generation

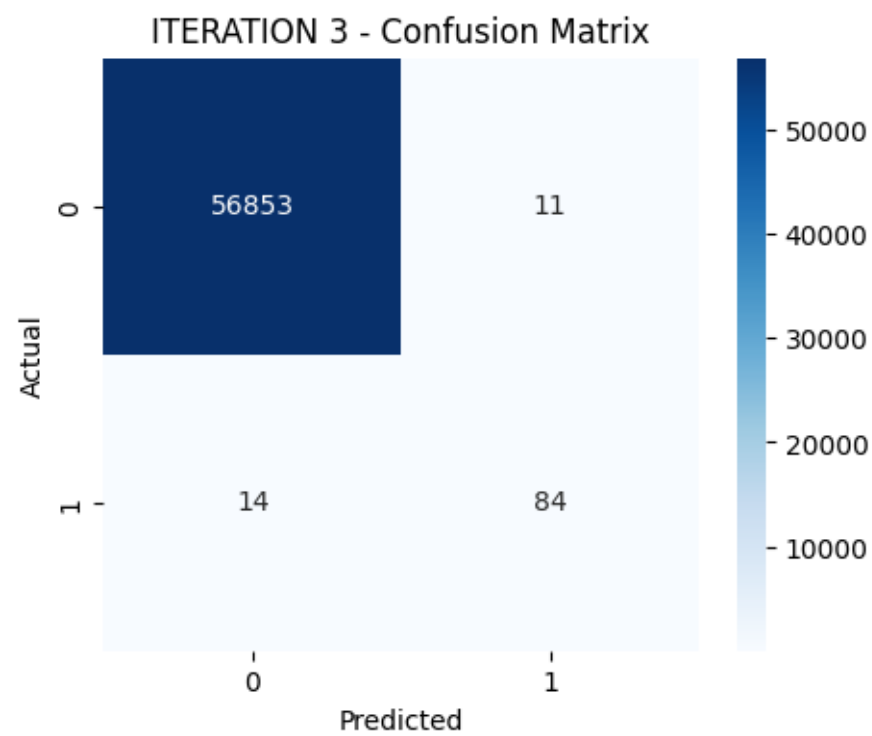
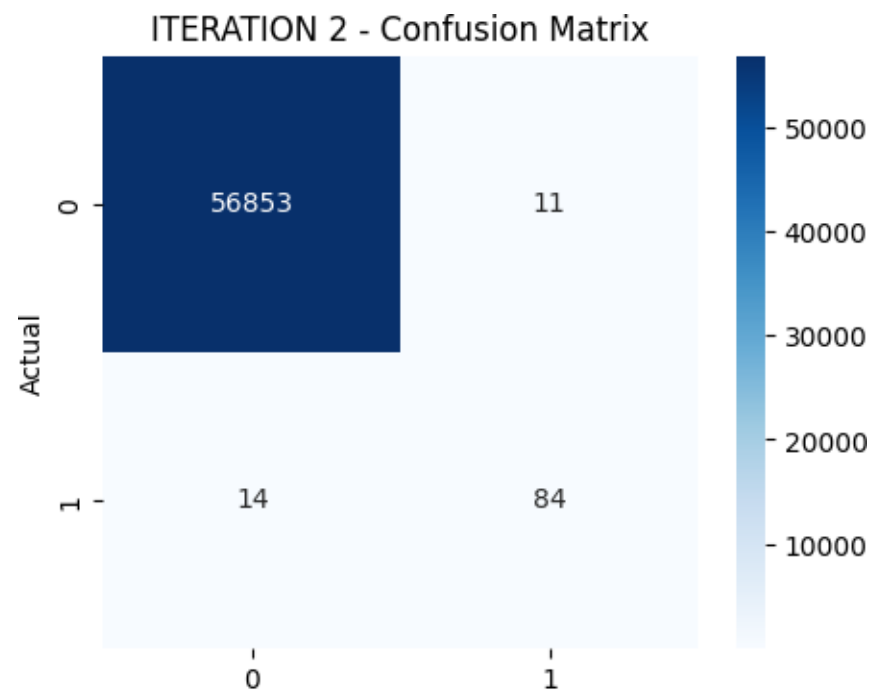
The **Gaussian Copula Synthesizer** from the SDV library generates synthetic samples: - Focused on minority class and error samples - Ensures the model learns from its mistakes

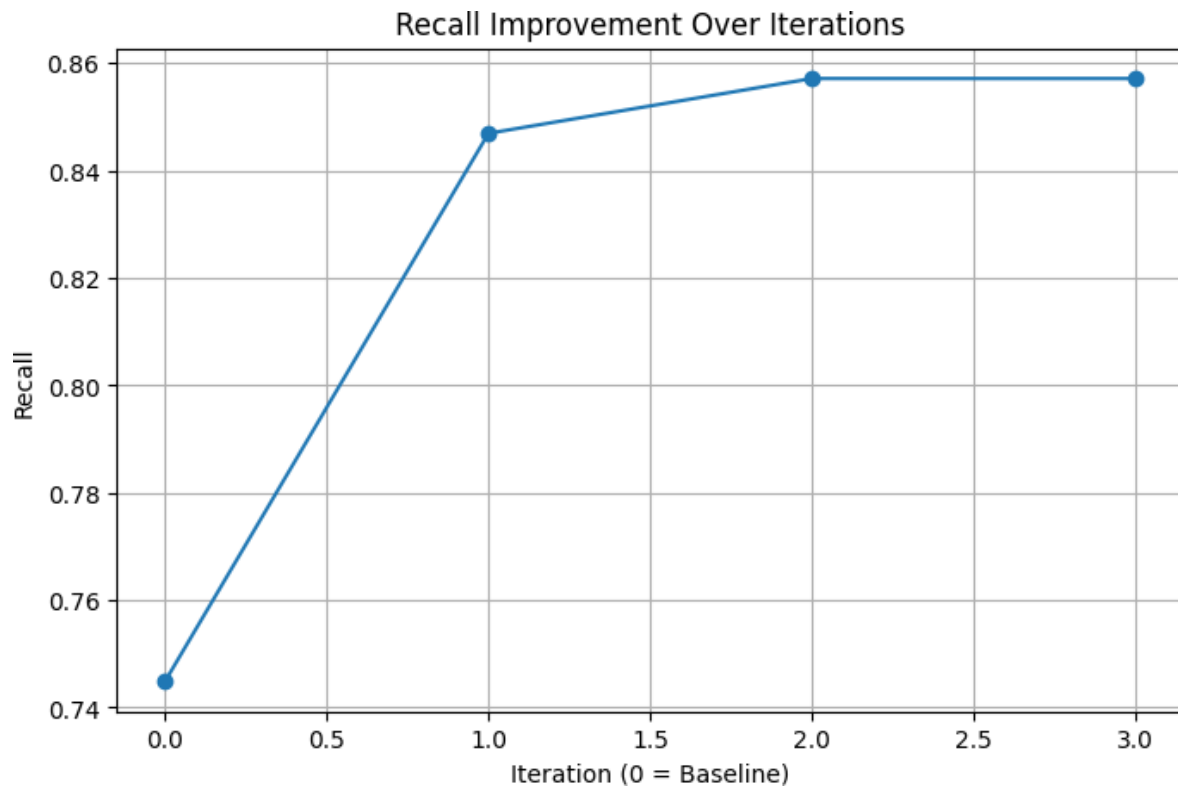
4.4 Adaptive Closed-Loop System

The iterative pipeline includes: 1. Train model 2. Identify false negatives 3. Generate targeted synthetic data 4. Augment training dataset 5. Retrain model 6. Evaluate performance 7. Repeat until improvement plateaus

5. Results:







5.1 Baseline Performance

- Recall (Class 1): 0.7449

5.2 Adaptive Augmentation Performance

- Final Recall (Class 1): 0.8571
- Total Improvement: **11.22%**

5.3 Observations

- Significant reduction in false negatives
 - Enhanced detection of minority class
 - Closed-loop adaptation successfully improves performance
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6. Challenges and Solutions:

1. **No initial improvement:** Synthetic data generation restricted to minority class and errors.
2. **Model bias toward majority class:** Used `class_weight='balanced'`.
3. **Overfitting risk:** Controlled augmentation and early stopping applied.

7. Conclusion:

The project demonstrates that adaptive synthetic augmentation can significantly improve model performance on imbalanced datasets. Key achievements include: - 11.22% improvement in minority class recall - Effective closed-loop augmentation system - Demonstrated applicability on a real-world dataset

8. Future Work:

- Explore GANs or VAEs for more complex augmentation
 - Incorporate fairness and privacy metrics
 - Develop a user-friendly interface for the toolkit
 - Apply the approach to healthcare and IoT datasets
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9. Tools and Technologies:

- Python, Scikit-learn, SDV
 - Pandas, NumPy
 - Matplotlib, Seaborn
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10. Acknowledgment:

This project provided practical experience in machine learning, handling imbalanced datasets, and synthetic data generation, contributing to both academic learning and professional development.
